SCORE/XX:

Math 304
Linear Algebra
November 22, 2017

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# PRACTICE MIDTERM III

Please show your work. Calculators and notes are forbidden. Only 1 brain (your own) allowed per exam. Please also note that some of the problems have multiple parts.

**0:** (Thanks to Souvik Gouswami for this problem.)
(a) Consider the vector space $P_3$, consisting of polynomials, in one variable, with degree $\leq 3$ and real coefficients. Please determine whether the polynomial $x^3+6x+4$ is in the span of $\{x^3+x^2+x, x^3+2x-6, x^2-5\}$.

Done in class on Tuesday, Nov. 28, 2017.

(b) Consider the vector space $\mathbb{R}^{2\times 2}$, and the subspace $W$ consisting of all matrices of the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a+b=c$, $b+c=d$, and $c+d=a$. What is the dimension of $W$?
**Hint:** Both problems can be easily converted into questions about subspaces in $\mathbb{R}^4$.

Done in class on Tuesday, Nov. 28, 2017

Instructor: J. Maurice Rojas

**1:** • Suppose $a$ is any column vector in $\mathbb{R}^n$ **not** equal to the zero vector.

**(a):** Please show that $L(x) := \frac{aa^T}{a^Ta}x$ is the map which projects any vector $x$ orthogonally onto the line generated by $a$. In other words, please show that (1) $L(\alpha a) = \alpha a$ for all $\alpha \in \mathbb{R}$, and (2) $L(x) = \vec{0}$ if $x$ is orthogonal to $a$.

(1) $L(\alpha a) = \frac{aa^T}{a^Ta}(\alpha a) = \frac{\alpha}{\cancel{a^Ta}} a\cancel{(a^Ta)} = \alpha a$. (Note that $aa^T \in \mathbb{R}^{n \times n}$ while $a^Ta$ is a real scalar.)

(2) $x$ orthogonal to $a$ means $x^Ta = a^Tx = 0$. So then, $x$ orthogonal to $a$ implies that $L(x) = \frac{aa^T}{a^Ta}x = \frac{a}{a^Ta}(a^Tx) = \frac{a}{a^Ta}(0) = \vec{0}$.

**(b):** Let $R(x) := x - 2L(x)$. ($R$ is known as a **Householder Reflection**, and is quite important in computing the $QR$ factorization in a numerically stable manner.) Please show that $R$ is the reflection which flips vectors across the hyperplane $H_a := \{x \in \mathbb{R}^n \mid a^Tx = 0\}$. In other words, please show that (1) $R(x) = x$ for all $x \in H_a$, and (2) $R(\alpha a) = -\alpha a$ for all $\alpha \in \mathbb{R}$.

(1) If $x \in H_a$ then, by definition, $a^Tx = 0$.

So then, $R(x) = x - 2L(x) = x - 2\frac{aa^T}{a^Ta}x = x - 2\frac{a}{a^Ta}(a^Tx)$

$= x - 2\frac{a}{a^Ta}(0)$

$= x - \vec{0} = x.$

(2) $R(\alpha a) = \alpha a - 2L(\alpha a) = \alpha a - 2\frac{aa^T}{a^Ta}(\alpha a) = \alpha\left(a - 2\frac{a}{\cancel{a^Ta}}\cancel{a^Ta}\right)$

$= \alpha(a - 2a)$

$= -\alpha a.$

**3:** Please find a basis for the subspace of $\mathbb{R}^4$ perpendicular to the span of $\begin{bmatrix}1\\1\\1\\3\end{bmatrix}$ and $\begin{bmatrix}1\\1\\1\\1\end{bmatrix}$.

This is the same as finding a basis for the solutions

$x = \begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix} \in \mathbb{R}^4$ of $\begin{bmatrix}1 & 1 & 1 & 1\\1 & 1 & 1 & 3\end{bmatrix} x = \begin{bmatrix}0\\0\end{bmatrix}$.

In other words, we are simply looking for a basis of the right nullspace of a matrix. Doing row reduction on our last $2\times 4$ matrix, we get

$$\begin{array}{c} \\ \tfrac{1}{2} \end{array}\begin{bmatrix}1 & 1 & 1 & 1\\0 & 0 & 0 & 2\end{bmatrix} \longrightarrow \begin{bmatrix}1 & 1 & 1 & 1\\0 & 0 & 0 & 1\end{bmatrix} \rightarrow \begin{bmatrix}\underline{1} & 1 & 1 & 0\\0 & 0 & 0 & \underline{1}\end{bmatrix}.$$

The last matrix is in reduced row echelon form, from which we can read off that columns $\underline{2}$ and $\underline{3}$ are free. So we easily get

$$\boxed{\left\{\begin{bmatrix}-1\\1\\0\\0\end{bmatrix}, \begin{bmatrix}-1\\0\\1\\0\end{bmatrix}\right\}}$$

as our desired basis.

**2:** The following algorithm takes an input set of column vectors $a_1, \ldots, a_n \in \mathbb{R}^n$ and produces a new set of column vectors $q_1, \ldots, q_n \in \mathbb{R}^n$, along with a set of numbers $\{r_{ij}\}$:

0. Let $r_{ij} = 0$ for all $i > j$.

```
FOR i = 1 TO n
    q_i = a_i
    FOR j = 1 TO i - 1
        r_ji = q_j^T q_i
        q_i = q_i - r_ji q_j
    END FOR
    r_ii = ||q_i||
    IF r_ii = 0
        STOP  /* We just discovered that the a_i were linearly dependent. */
    END IF
    q_i = q_i / r_ii
END FOR
```

**(a):** Please execute the above algorithm for the following inputs: $n = 3$, $a_1 := \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$, $a_2 := \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$, $a_3 := \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$.

(A slight variation was done in class. So, for the sake of practice, I'll present a solution to <u>this</u> problem as stated.)

Following the loops literally, we obtain:

<u>$i=1$</u>: $q_1 = a_1 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$, inner loop empty! (since $i=1 \Rightarrow$ inner loop has $j$ ranging from 1 to 0, which makes no sense!) $\Rightarrow r_{11} = \sqrt{1+2^2+4^2} = \sqrt{21}$. $\Rightarrow q_1 = \begin{bmatrix} 1/\sqrt{21} \\ 2/\sqrt{21} \\ 4/\sqrt{21} \end{bmatrix}$.

<u>$i=2$</u>: $q_2 = a_2 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$, <u>$j=1$</u>, $r_{12} = q_1^T q_2 = [\frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}, \frac{4}{\sqrt{21}}] \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = -\frac{7}{\sqrt{21}}$, $q_2 = q_2 - (-\frac{7}{\sqrt{21}}) \begin{bmatrix} 1/\sqrt{21} \\ 2/\sqrt{21} \\ 4/\sqrt{21} \end{bmatrix} = \begin{bmatrix} 4/3 \\ 2/3 \\ -2/3 \end{bmatrix} \Rightarrow r_{22} = \frac{2}{3}\sqrt{6} \Rightarrow q_2 = \frac{q_2}{r_{22}} = \begin{bmatrix} 2/\sqrt{6} \\ 1/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix}$

<u>$i=3$</u>: $q_3 = a_3 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$, <u>$j=1$</u>, $r_{13} = q_1^T q_3 = [\frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}, \frac{4}{\sqrt{21}}] \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} = \frac{8}{\sqrt{21}}$, $q_3 = q_3 - \frac{8}{\sqrt{21}} \begin{bmatrix} 1/\sqrt{21} \\ 2/\sqrt{21} \\ 4/\sqrt{21} \end{bmatrix} = \begin{bmatrix} -8/21 \\ 26/21 \\ -11/21 \end{bmatrix}$.

<u>$j=2$</u>, $r_{23} = q_2^T q_3 = [\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}] \begin{bmatrix} -8/21 \\ 26/21 \\ -11/21 \end{bmatrix} = \frac{1}{\sqrt{6}}$, $q_3 = q_3 - \frac{1}{\sqrt{6}} \begin{bmatrix} 2/\sqrt{6} \\ 1/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix} \cdot \frac{5}{14} \Rightarrow r_{33} = \frac{5}{14}\sqrt{14} \Rightarrow q_3 = \begin{bmatrix} -2/\sqrt{14} \\ 3/\sqrt{14} \\ -1/\sqrt{14} \end{bmatrix}$.

**(b):** Please find the $QR$ factorization of $A$.

Assembling all the entries from our calculations above, we get:

$$\underbrace{\begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 2 \\ 4 & -2 & 1 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1/\sqrt{21} & 2/\sqrt{6} & -2/\sqrt{14} \\ 2/\sqrt{21} & 1/\sqrt{6} & 3/\sqrt{14} \\ 4/\sqrt{21} & -1/\sqrt{6} & -1/\sqrt{14} \end{bmatrix}}_{Q} \cdot \underbrace{\begin{bmatrix} \sqrt{21} & -7/\sqrt{21} & 8/\sqrt{21} \\ 0 & \frac{2}{3}\sqrt{6} & 1/\sqrt{6} \\ 0 & 0 & \frac{5}{14}\sqrt{14} \end{bmatrix}}_{R}$$

**5:** Suppose the following information is known about a matrix $A$:

$$A\begin{bmatrix}1\\2\\4\end{bmatrix}=9\begin{bmatrix}1\\1\\1\end{bmatrix},\ A\begin{bmatrix}1\\-3\\9\end{bmatrix}=2\begin{bmatrix}1\\-3\\9\end{bmatrix},\ \text{and } A\begin{bmatrix}1\\1\\1\end{bmatrix}=3\begin{bmatrix}1\\1\\1\end{bmatrix}.$$

**(a):** Please find the eigenvalues of $A$.

The 2nd equality clearly implies the eigen-pair $\left(2, \begin{bmatrix}1\\-3\\9\end{bmatrix}\right)$.

The 3rd equality clearly implies the eigenpair $\left(3, \begin{bmatrix}1\\1\\1\end{bmatrix}\right)$.

The right-hand sides of the 1st & 3rd equalities are proportional, from which we get $A\left(\begin{bmatrix}1\\2\\4\end{bmatrix}-3\begin{bmatrix}1\\1\\1\end{bmatrix}\right)=9\begin{bmatrix}1\\1\\1\end{bmatrix}-3\cdot3\cdot\begin{bmatrix}1\\1\\1\end{bmatrix}=\vec{0}$.

So we obtain the third eigenpair $\left(0, \begin{bmatrix}-2\\-1\\1\end{bmatrix}\right)$.

**(b):** Please find the eigenvectors of $A$.

As noted in (a), they are (nonzero) multiples of

$$\begin{bmatrix}1\\-3\\9\end{bmatrix},\ \begin{bmatrix}1\\1\\1\end{bmatrix},\ \text{or } \begin{bmatrix}-2\\-1\\1\end{bmatrix}.$$

**(c):** Is $A$ invertible? Please explain...

No, because invertible matrices have nonzero determinant. Our $A$ has a 0 eigenvalue, implying that the constant term of the characteristic polynomial vanishes, i.e., $\det A = 0$.

**(d):** Is $A$ diagonalizable? Please explain...

Yes, because we have 3 linearly independent eigenvectors, and $A$ is $3\times3$. So then, our preceding observations imply

$$A\begin{bmatrix}1&1&-2\\-3&1&-1\\9&1&1\end{bmatrix}=\begin{bmatrix}1&1&-2\\-3&1&-1\\9&1&1\end{bmatrix}\begin{bmatrix}2&0&0\\0&3&0\\0&0&0\end{bmatrix}.$$

So then we get $A=\begin{bmatrix}1&1&-2\\-3&1&-1\\9&1&1\end{bmatrix}\begin{bmatrix}2&&\\&3&\\&&0\end{bmatrix}\begin{bmatrix}1&1&-2\\-3&1&-1\\9&1&1\end{bmatrix}^{-1}$.

**4:** The following information is known about a matrix $A$:

$$A\begin{bmatrix}1\\2\\3\\4\end{bmatrix}=\begin{bmatrix}1\\1\end{bmatrix} \text{ and } A\begin{bmatrix}2\\3\\4\\1\end{bmatrix}=\begin{bmatrix}2\\2\end{bmatrix}.$$

Also, every column of $A$ is a multiple of $\begin{bmatrix}1\\1\end{bmatrix}$.

(a) What is the dimension of the right-nullspace of $A$? Please explain why.

First observe that $A$ must be $2\times 4$. From the last sentence, we know that $\dim Col(A) \leq 1$. From the first given equality, we see that $Col(A) \neq \{\vec{0}\}$. So $\dim Col(A) = 1$. By the 4-Subspaces Theorem, $\dim Row(A) = 1$ as well, and $\dim Row(A) + \dim RightNull(A) = \#$ Columns of $A$. So we get $1 + \dim RightNull(A) = 4 \Rightarrow$ our desired dimension is $\boxed{3}$

(b) What is the dimension of the row space of $A$? Please explain why.

From our observations in Part (a), we already observed that $\dim Col(A) = \dim Row(A) = \boxed{1}$

(c) What is the dimension of the left-nullspace of $A$? Please explain why.

From the 4-Subspaces Theorem, we have

$\dim Col(A) + \dim LeftNull(A) = \#$ Rows of $A$. So we get

$1 + \dim LeftNull(A) = 2$

and thus $\dim LeftNull(A) = \boxed{1}$.

**6:** Find an orthonormal basis for the subspace of $\mathbb{R}^5$ generated by the following pair of vectors:

$$\begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

We can actually perform a slight variant of Gram-Schmidt:

(1) Renormalize $a_1 := \begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \\ 1 \end{bmatrix}$: let $q_1 := \frac{a_1}{|a_1|} = \begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \\ 1 \end{bmatrix} \Big/ \sqrt{1^2+3^2+1^2+2^2+1^2}$

$$= \begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \\ 1 \end{bmatrix} \Big/ 4.$$

(2) let $q_2 := a_2 := \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$ and $r_{12} := q_1^T q_2 = \frac{1\cdot 1 + 3\cdot 2 + (-1)(1) + 2\cdot 0 + 1\cdot 1}{4} = \frac{1+6-1+1}{4} = \frac{7}{4}$,

and straighten out $q_2$ by subtracting a multiple of $q_1$:

$$q_2 := q_2 - r_{12} q_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 1 \end{bmatrix} - \frac{7}{4} \begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \\ 1 \end{bmatrix} \Big/ 4$$

$$= \begin{bmatrix} 9 \\ 11 \\ 23 \\ -14 \\ 9 \end{bmatrix} \Big/ 16.$$

(3) let $r_{22} := |q_2|$ and renormalize $q_2$:

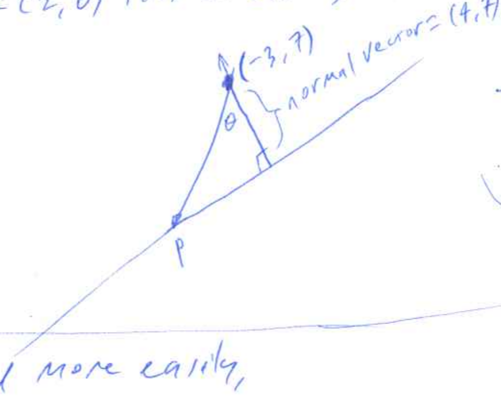
$$q_2 := \frac{q_2}{r_{22}} = \left( \begin{bmatrix} 9 \\ 11 \\ 23 \\ -14 \\ 9 \end{bmatrix} \Big/ 16 \right) \Big/ \left( \frac{1}{16} \sqrt{9^2+11^2+23^2+(-14)^2+9^2} \right)$$

$$= \begin{bmatrix} 9 \\ 11 \\ 23 \\ -14 \\ 9 \end{bmatrix} \Big/ 12\sqrt{7} = \begin{bmatrix} 3/4 \\ 11/12 \\ 23/12 \\ -7/6 \\ 3/4 \end{bmatrix} \Big/ \sqrt{7}.$$

Our $\{q_1, q_2\}$ then yield our desired orthonormal basis.

7: (a) Is $(-3,7)$ closer to the line described by $4x+7y=8$ or the line described by $-5x+3y=0$?

Let $L_1$ and $L_2$ be our respective lines.
To compute the distance between $(-3,7)$ and $L_1$, we'll first need a point on $L_1$.
Clearly, $p=(2,0)$ lies on $L_1$. So then, we can compute the distance as in class:




$\Rightarrow$ Distance from $(-3,7)$ to $L_1$

$$= |p-(-3,7)|\,|\cos\theta| = |(2,0)-(-3,7)|\,|\cos\theta|$$

$$= |(5,-7)| \left| \frac{(p-(-3,7))\cdot(4,7)}{|p-(-3,7)|\,|(4,7)|} \right|$$

$$= \cancel{|(5,-7)|} \frac{|(5,-7)\cdot(4,7)|}{\cancel{|(5,-7)|}\,|(4,7)|}$$

$$= \frac{|(5,-7)\cdot(4,7)|}{\sqrt{4^2+7^2}} = \frac{29}{\sqrt{65}}$$

Similarly, and more easily, the distance between $(-3,7)$ and $L_2$ is $\dfrac{|(-3,7)\cdot(-5,3)|}{|(-5,3)|} = \dfrac{36}{\sqrt{34}}$.

So to compare, we can square:

$$\underbrace{\frac{36^2}{34}}_{L_2} \text{ vs. } \underbrace{\frac{29^2}{65}}_{L_1} \Rightarrow 36^2\cdot 65 \text{ vs. } 29^2\cdot 34$$

$\Downarrow$

84240 vs. 28594

$\Rightarrow$ $L_1$ is closer

(b) Which of the following points is closer to the plane described by $x-2y-4z=0$? $(9,10,10)$ or $(10,9,10)$?

Similar to above, we merely compare

$$\frac{|(9,10,10)\cdot(1,-2,-4)|}{|(1,-2,-4)|} \text{ vs. } \frac{|(10,9,10)\cdot(1,-2,-4)|}{|(1,-2,4)|}$$

or, equivalently,

$$|(9,10,10)\cdot(1,-2,-4)| \text{ vs. } |(10,9,10)\cdot(1,-2,-4)|$$

or

$$|9-20-40| \text{ vs. } |10-2\cdot 9-4\cdot 10|$$

or

$$51 \text{ vs. } 48$$

$\Rightarrow$ $(10,9,10)$ is closer.